

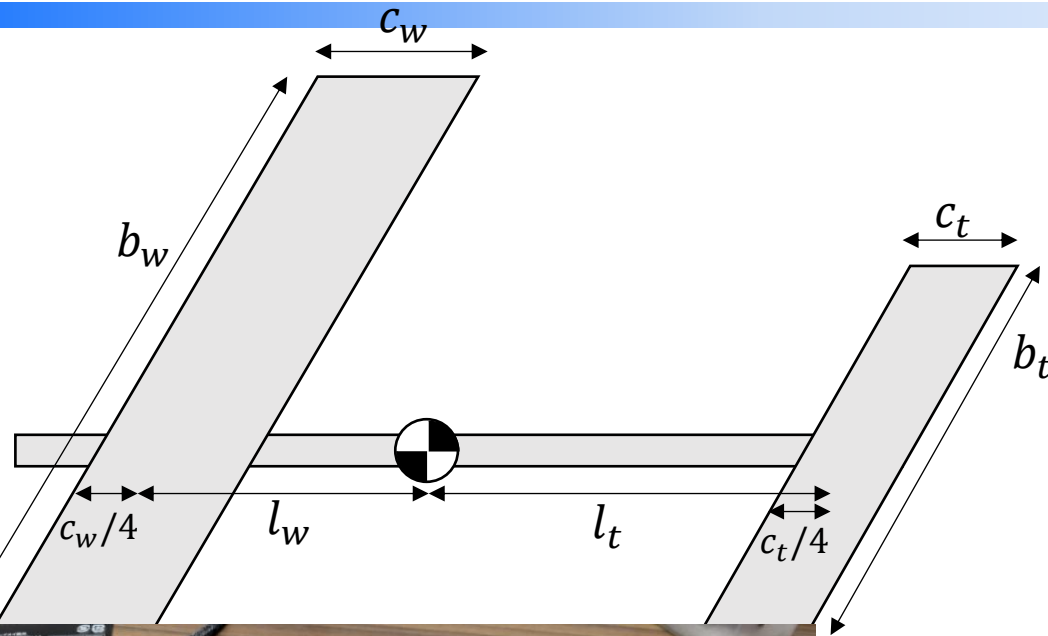
Aeronautical Systems B Lecture: 6

航空宇宙システム特論B 第6回

1. Build a glider and perform vertical motion simulations. Stability Analysis

1. グライダーを製作し縦運動のシミュレーションを行う

Example



Styrene Sheet, 1mm thick(Main wing, tale wings)

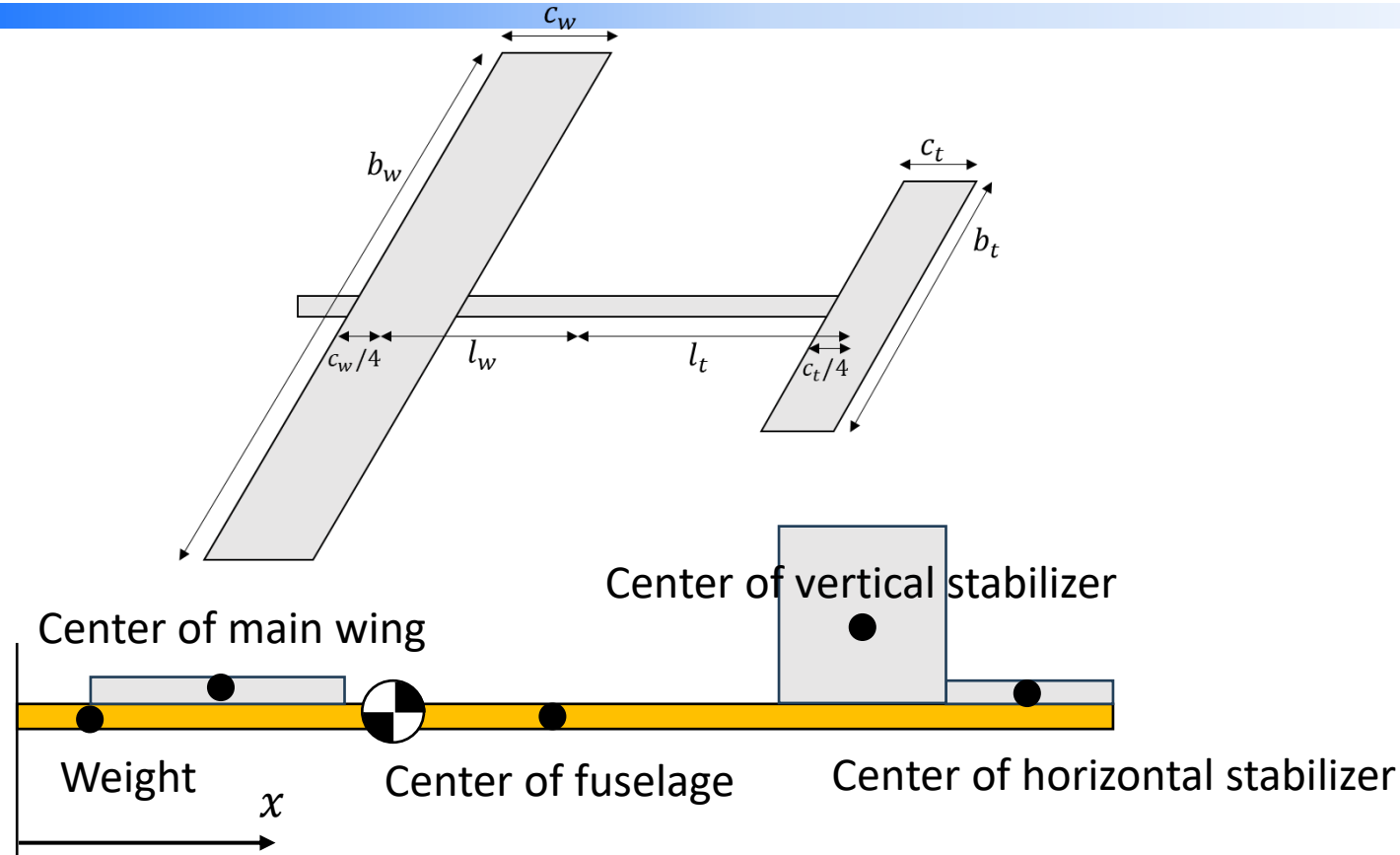
Surface density :0.0098 g/cm²

3mm x 3mm balsa wood stick(Fuselage)

Line density :0.0022 g/cm

Weight :1.05 g

Example

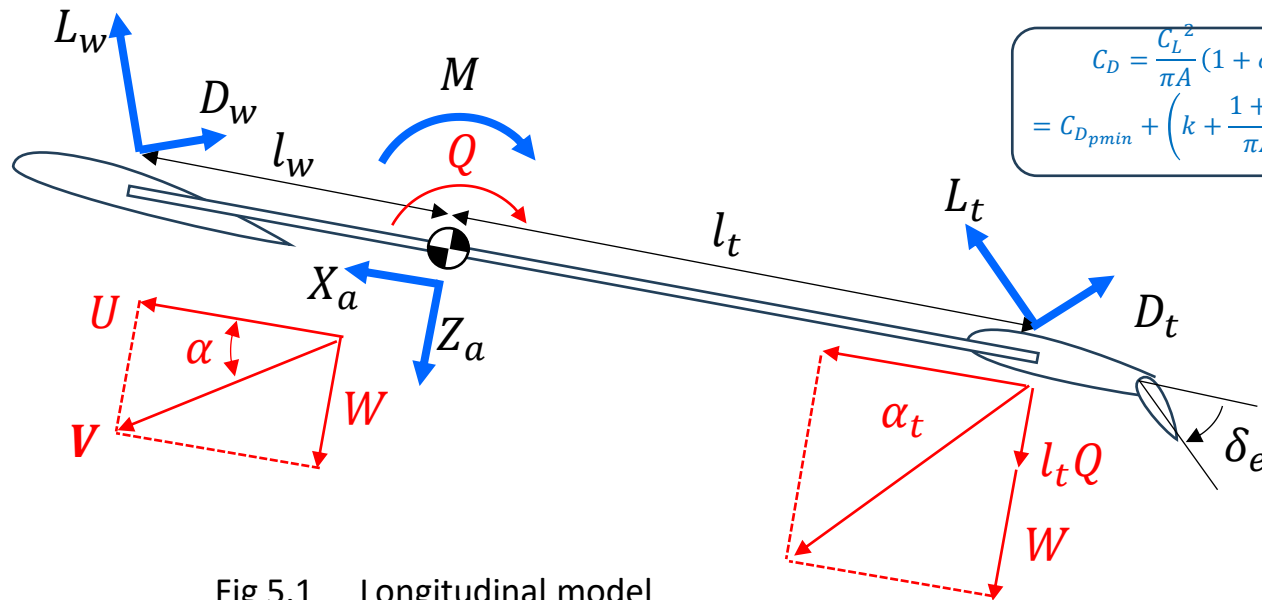


Calculate the center of gravity and moment of inertia from the mass and position of each element.

Center of gravity :
$$\sum_i (x_i - x_{cg})m_i = 0 \quad (\text{B6-1})$$

Moment of inertia :
$$I_{yy} = \sum_i (x_i - x_{cg})^2 m_i \quad (\text{B6-2})$$

Effect of Control Surfaces



$$C_D = \frac{C_L^2}{\pi A} (1 + \delta) + C_{D_{pmin}} + k C_L^2$$

$$= C_{D_{pmin}} + \left(k + \frac{1 + \delta}{\pi A} \right) C_L^2 = C_{D_{pmin}} + \frac{C_L^2}{\pi e A} \quad (B5-30)$$

Fig 5.1 Longitudinal model

Angle of attack of main wing

$$\alpha_w \cong \alpha$$

$$= \tan^{-1} \frac{W}{U} \quad (B5-6a)$$

Angle of attack of horizontal stabilizer

$$\alpha_t = \tan^{-1} \frac{W + l_t Q}{U} \quad (B5-6b)$$

Lift/Drag of main wing

$$L_w = \frac{1}{2} \rho V^2 S_w C_{L_w} \quad (B5-7a)$$

$$D_w = \frac{1}{2} \rho V^2 S_w C_{D_w} \quad (B5-7b)$$

Lift/Drag of horizontal stabilizer

$$L_t = \frac{1}{2} \rho V^2 S_t C_{L_t} \quad (B5-7c)$$

$$D_t = \frac{1}{2} \rho V^2 S_t C_{D_t} \quad (B5-7d)$$

Lift/Drag coefficient of main wing

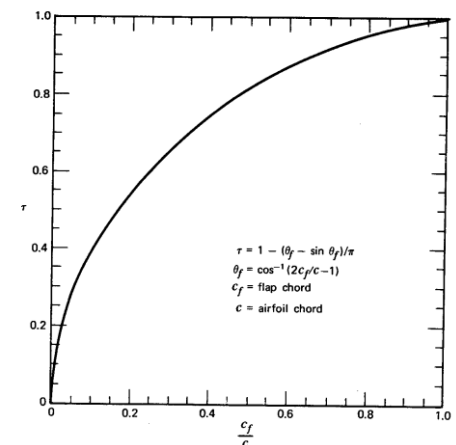
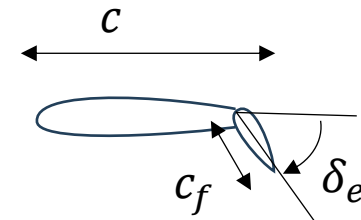
$$C_{L_w} = C_{L_{w0}} + a_w \alpha_w \quad (B5-8a)$$

$$C_{D_w} = C_{D_{w0}} + \frac{C_{L_w}^2}{\pi e_w A_w} \quad (B5-8b)$$

Lift/Drag coefficient of horizontal stabilizer

$$C_{L_t} = C_{L_{t0}} + a_t (\alpha_t + \tau \delta_e) \quad (B6-3)$$

$$C_{D_t} = C_{D_{t0}} + \frac{C_{L_t}^2}{\pi e_t A_t} \quad (B5-8d)$$



1. Build a glider using styrene sheet, balsa wood rods, and weights.
2. Measure the specifications of the constructed aircraft.
3. Determine the center of gravity position and moment of inertia((B6-1) and (B6-2)).
4. Express the aerodynamic forces X_a and Z_a acting on the aircraft in terms of the aircraft state variables using equations (B5-3) to (B5-15) and (B6-3).
5. Implement the aerodynamic force model in the Matlab program.
6. Conduct simulations for various initial values.